

Verificare metoda Kramers

$$h := 6.625 \cdot 10^{-34} \quad m_e := 9.1 \cdot 10^{-31} \quad e_e := 1.6 \cdot 10^{-19}$$

$$h_b := \frac{h}{2 \cdot \pi}$$

Definim pasul de discretizare pe axa x, in Angstromi

$$\delta x := 10^{-10}$$

Definirea potentialului

$$V_0 := 0.2$$

Potential rectangular

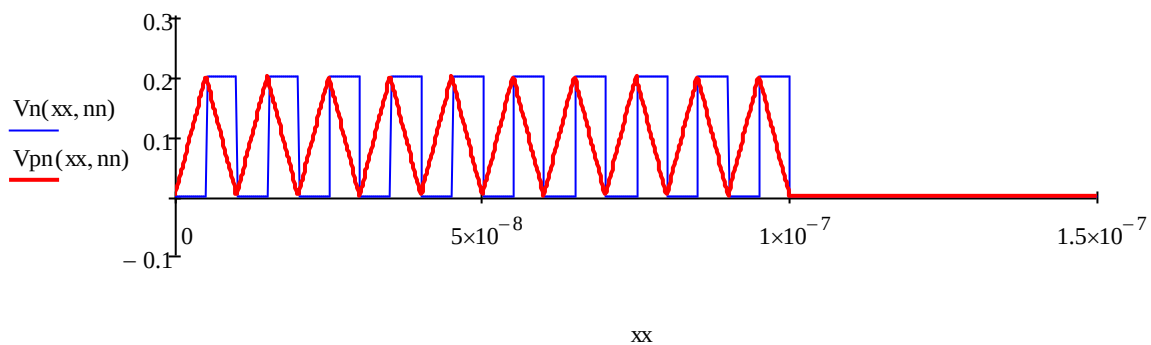
Potential triunghiular

$$V_n(x, n) := \begin{cases} 0 & \\ V_0 & \text{if } \text{mod}(x, 100 \cdot \delta x) \geq 50 \cdot \delta x \\ 0 & \text{if } x > n \cdot 100 \cdot \delta x \end{cases}$$

$$V_{pn}(x, n) := \begin{cases} \text{aux} \leftarrow \text{mod}(x, 100 \cdot \delta x) \\ \text{aux} \leftarrow 100 \cdot \delta x - \text{aux} & \text{if } \text{aux} > 50 \cdot \delta x \\ \text{out} \leftarrow V_0 \left(\frac{2 \cdot \text{aux}}{100 \cdot \delta x} \right) \\ \text{out} \leftarrow 0 & \text{if } x > n \cdot 100 \cdot \delta x \\ \text{out} \end{cases}$$

Numarul de celule elementare

$$n_n := 10$$



$x_{\min} := 0$ $N := 1500$

$i := 0..N$

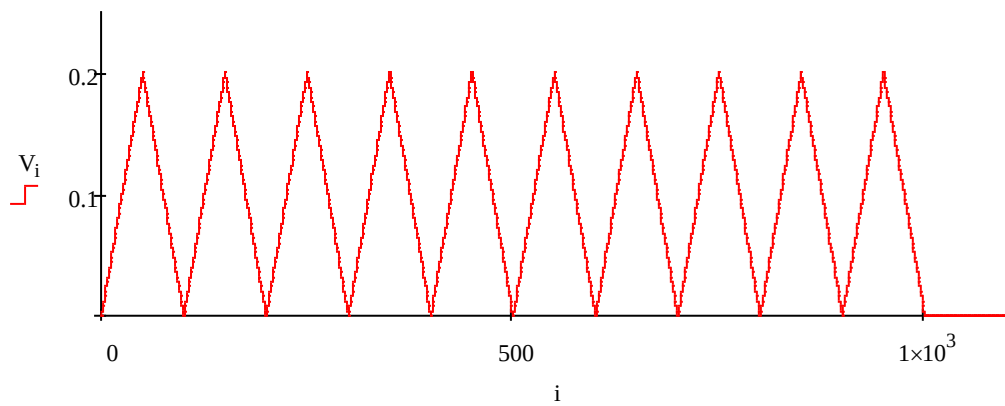
$x_i := x_{\min} + \delta x \cdot i$

In acest exemplu consideram potentialul triunghiular

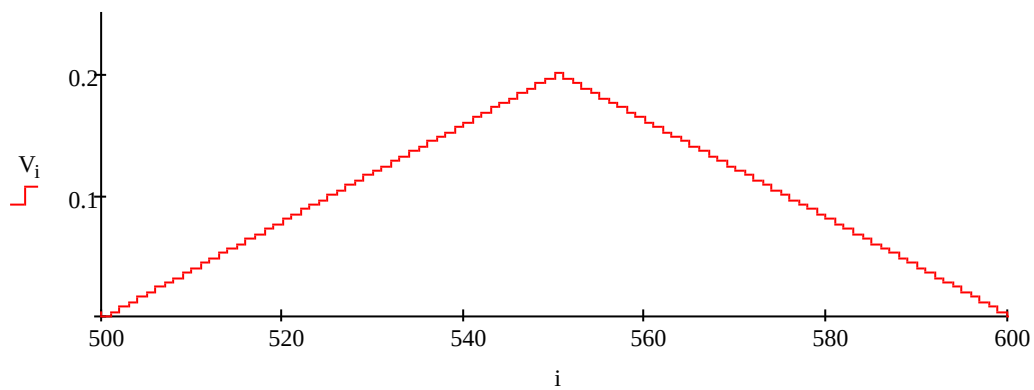
$V_i := V_{pn}(x_i, m)$

Consideram masa efectiva a electronului in GaAs

$m = 0.063 m_e$



Detaliu al graficului pe o celula elementara. Se observa aproximarea cu potentiale constante pe portiuni



Vectorul de unda:

$$\text{const} := \frac{2 \cdot m \cdot e e}{\hbar^2}$$

$$k(i, E) := \sqrt{\text{const} \cdot (E - V_i)}$$

Matricea de transfer

$$\text{R}(i, E) := \frac{k(i, E)}{k(i - 1, E)}$$

$$\text{Mtransfer}(i, E) := \frac{1}{2} \cdot \begin{pmatrix} 1 + R(i, E) & 1 - R(i, E) \\ 1 - R(i, E) & 1 + R(i, E) \end{pmatrix}$$

$$\text{Mtransport}(i, E) := \begin{pmatrix} e^{-i k(i, E) \cdot \delta x} & 0 \\ 0 & e^{i k(i, E) \cdot \delta x} \end{pmatrix}$$

$$M_i(i, E) := \text{Mtransfer}(i, E) \cdot \text{Mtransport}(i, E)$$

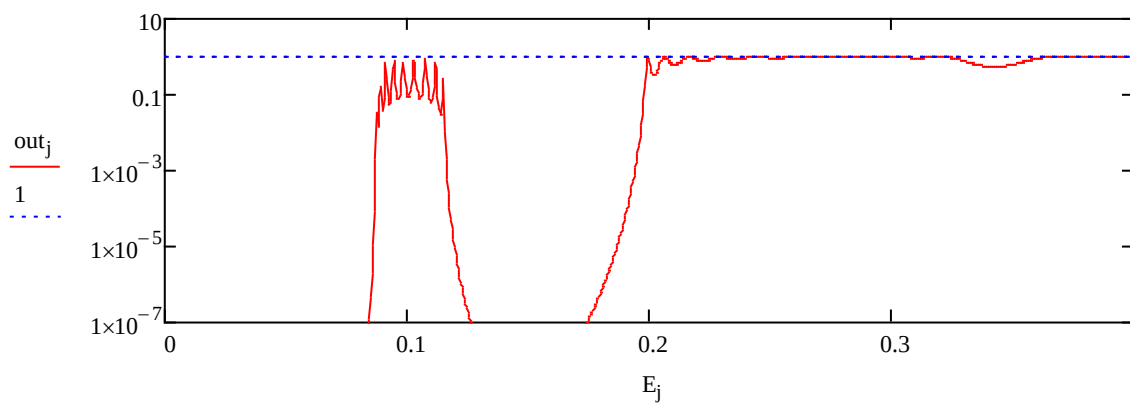
$$M(E) := \left(\prod_{i=1}^{N-1} M_i(i, E) \right) \cdot \text{Mtransfer}(N, E)$$

$$\text{T}(E) := \frac{k(N, E)}{k(0, E)} \cdot \left(\left| M(E)_{0,0} \right| \right)^{-2}$$

$$j := 0..1000$$

$$E_j := 0.00000000001 + j \cdot \frac{10^{-3}}{2} \quad \text{max}(E) = 0.5$$

$$\text{out}_j := T(E_j)$$



Aplicam conditia Kramers

Definim numarul de puncte din celula elementara

$$\text{NCE} := 101$$

$$\text{MCE}(E) := \left(\prod_{i=1}^{\text{NCE}-1} \text{Mi}(i, E) \right) \cdot \text{Mtransfer}(\text{NCE}, E)$$

$$\text{TR}(E) := \frac{1}{2} \cdot |\text{tr}(\text{MCE}(E))|$$

